

SUMUKH M

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Profile

Computer Science Student at PES University | Passionate about Software Development and AI | Seeking Internship Opportunities

PROJECTS

AI PIPELINES WITH LANGCHAIN & RAG | PYTHON, LANGCHAIN, VECTOR STORES | 2025 - 2026

- Built a series of Python projects exploring the LangChain framework, covering chat models, prompt templates, structured output, and output parsers.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using document loaders, text splitters, vector stores, and retrievers to answer queries over custom documents.

SDN PACKET DROP SIMULATOR | PYTHON, MININET, POX, OPENFLOW | 2025

- Built a Software-Defined Networking (SDN) simulator to model centralized traffic control using Mininet and a POX controller.
- Implemented selective packet-drop rules by inspecting source/destination IPs and protocols at the controller, replicating firewall-like behavior over OpenFlow.

MINES GAME WITH RIGGED MECHANIC | HTML, CSS, JAVASCRIPT | 2025

- Developed a browser-based Minesweeper-style game with a custom rigged gameplay mechanic and toggleable fairness switch.
- Implemented game logic using JavaScript event handling and conditional algorithms to control mine placement and user interactions.

Education

B.TECH. in PES UNIVERSITY | June 2024-2028 | Electronic City, Bangalore, Karnataka, India

Skills & Abilities

Languages : C, Python, JavaScript, C++

Web Development : HTML, CSS, React.js, Node.js, MongoDB (MERN Stack)

AI / ML : LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, Prompt Engineering

Networking : Software-Defined Networking (SDN), Mininet, POX Controller, OpenFlow Protocol

Core CS : Data Structures & Algorithms, Operating Systems, Computer Networks

Tools : Git, GitHub, Linux, VS Code

Activities and Interests

CTF Challenges, Cybersecurity, Machine Learning & AI, LangChain/RAG Experimentation, Hackathons,

